

PDF Utility Pro

An offline PDF toolkit for Windows — split, merge, organise, redact and edit PDF files without any document ever leaving the computer.

Built as an ICAI Capstone Project.

100% Offline — Your Documents Never Leave Your Computer. No cloud service, no upload, no external API.

The application opens no network connection and works with the network cable unplugged.

Folder structure



PDFUtilityPro/

└─ main.py	Entry point — start the application here
└─ requirements.txt	Dependencies
└─ build_exe.bat	One-click Windows .exe build
└─ README.md	This file
└─ app/	
└─ theme.py	Seven colour themes + live theme switching
└─ settings.py	Local settings (theme, recent files)
└─ pdf_ops.py	PDF engine — no GUI, independently testable
└─ common.py	Shared widgets, page canvas, background worker
└─ main_window.py	Toolbar, theme picker, screen stack, Help
└─ pages/	
└─ home.py	Home screen, five function cards, theme picker
└─ split_page.py	1. Split (with batch mode)
└─ merge_page.py	2. Merge
└─ organise_page.py	3. Organise (single file + batch tabs)
└─ organise_batch.py	3b. Organise — batch mode
└─ redact_page.py	4. Redact
└─ edit_page.py	5. Edit
└─ docs/	
└─ PROJECT_REPORT.md	Full capstone documentation
└─ USER_MANUAL.md	How to use each screen
└─ TEST_CASES.md	Test plan and results
└─ DEMO_WORKFLOW.md	15-minute live demonstration script
└─ tools/	
└─ self_test.py	Automated engine tests (38 checks)
└─ gui_smoke_test.py	Builds every screen in every theme
└─ make_samples.py	Regenerates the demo PDFs
└─ samples/	Demo PDFs for the evaluation

Installation (development / demonstration machine)

Requires Python 3.10 or newer. Python 3.12 was used for development.



```
cd PDFUtilityPro
python -m venv .venv
.venv\Scripts\activate      (Windows)
source .venv/bin/activate   (Linux / macOS)
pip install -r requirements.txt
python main.py
```

Running the tests

```

python tools/self_test.py          # PDF engine, 38 checks
python tools/gui_smoke_test.py samples/Sample_Client_Report.pdf
```

`self_test.py` builds its own sample files, runs every operation and prints a PASS/FAIL line for each. It needs no interface and no internet.

Building the Windows .exe

Double-click `build_exe.bat`, or run:

```

pip install pyinstaller
pyinstaller --noconfirm --clean --name PDFUtilityPro --windowed --onefile ^
--hidden-import pymupdf --collect-submodules pymupdf main.py
```

The result is `dist\PDFUtilityPro.exe` — a single file that runs on any Windows PC without Python installed. First launch takes a few seconds while the one-file bundle unpacks.

The seven colour themes

Blue · Green · Purple · Orange · Red · Teal · Dark

Choose a theme from the dropdown in the toolbar, or from the swatches on the home screen. The change applies to every screen immediately — including custom-drawn surfaces such as the page canvas and the thumbnail grid — and is saved, so the application reopens in the same theme.

Themes are defined as colour tokens in `app/theme.py`. Adding an eighth theme means adding one `Theme(...)` entry and one line in `THEME_ORDER`; no other file needs to change.